

Aviral Khanna

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WORK EXPERIENCE

Razorpay Software Ltd. | Internship | Bengaluru, India

September 2025 – Present

- **AI & Automation Core Team:** Designed and deployed an internal **log-detection alert automation system** processing **10 crore+ logs/day**, delivering real-time insights and reducing incident-response latency by ~40% in live case scenarios.
- **Agentic AI Pipeline:** Built a fully automated workflow using **pidata(agno)**, **Coralogix**, **DevRev**, **Slack integrations**, and in-house tooling to orchestrate multi-step AI decision flows improving efficiency by 60%.
- **Codebase Resolution:** Diagnosed and fixed product's production bugs across critical Razorpay's payment products like: **core payments, routing, terminals, orders, webhooks, refunds, ledger and settlements** ensuring system reliability and customer satisfaction.

Founder and Technical Head | CCS | [Link](#) | [GitHub](#)

April 2023 - Present

Team of 34 students & 14 faculty | ₹1.5L+ funding | 1500+ users

Deployed Products:

- **Patent Project- DASES (Digital Answer Sheet Evaluation System):** An agentic AI solution for handwritten answer sheet evaluation using OCR, LaTeX parsing, rubric-based logic, and fine-tuned LLMs via LangGraph. Tested on 1500+ students across multiple disciplines including Law, CS, and Health Sciences. [Link](#)
- **Resume-JD Matcher:** An OCR-based ML/NLP pipeline that evaluates resume-job description compatibility, providing detailed feedback and match scores. Live and actively used by 500+ students to improve placement readiness. [Link](#)
- **Internship Evaluator:** A centralized portal used by the Career Services department to assess student performance during internships. Includes rubric management, progress tracking, and automated feedback generation. Piloted with 100+ students across domains.
- **UPES Custom Chatbot:** A university-specific AI chatbot built using neu4, graph databases, and embedding-based query learning to handle academic, administrative, and campus-related queries dynamically. Currently in active use for student support.
- Developed **LexiBot, SHODH Grant Recipient (₹49,000, 2025–26)**, an AI-powered legal assistant providing guidance for consumer, traffic, and women harassment cases. [Link](#)

AI/ML Software Developer Engineer Intern | VertxAI Labs | [Link](#) | [GitHub](#)

April 2025 - September 2025

- **Developed agentic AI models** for investor-founder and multimedia mapping using fine-tuned Gemini APIs, GPT-4, Claude, Whisper, achieving 96.54% input understanding accuracy via a full ML/NLP pipeline.
- **Built and deployed 4+ AI-driven tools**, including an email automation agent, investor-founder matching model, AI pitch deck generator, and pitch evaluator boosting workflow efficiency by 87%.
- **Served 500+ users** with stable, scalable performance across applications tailored for the start-up investment ecosystem.
- Improved system design and database integration using core concepts like caching and distributed systems, reducing monthly database call costs by over 20%. This performance gain led to an extended internship tenure and performance-based incentives.

ML Research Collaborator (with PhD scholar) | IIIT Delhi | [GitHub](#)

November 2023 – April 2024

- **Developed deep learning models** using Whisper, LLaMA3, CNNs, RNNs, LSTMs, and Transformers for speech-driven mental health assessment.
- **Achieved 90%+ accuracy** in autism detection and improved depression detection by 25% through continual learning on diverse datasets.
- **Enhanced model robustness** using adaptive techniques and inter-speaker variability handling in speech feature extraction.

Project Development Intern | IBM

June 2025– July 2025

- Collaborated in a team of 5 on deep learning models for vehicle automation, focusing on accuracy enhancement and system efficiency.
- Improved object detection, alert systems and scene understanding using YOLOv11, depth analysis, and custom finder algorithms, achieving accuracy of 98.3%.
- Benchmarked CNN, RNN, and Transformer architectures on a custom dataset of 15,000 images, achieving over 87% train-test accuracy.

Web Development & ML Engineer Intern | Hakencreuz Innovations

January 2025 – March 2025

- Optimized existing ML models to enhance software scalability and performance, improving inference speed using React and reducing resource usage across deployments.
- Developed intuitive front-end interfaces using MERN stack and integrated ML models into web applications.
- Collaborated with the team on brainstorming pipeline, technical discussions, and AI-driven product development.

ML Research Collaborator (with PhD scholar) | Ulster University (London) | [GitHub](#)

November 2023 – April 2024

- **Extracted speech features** such as MFCCs, spectrograms, and pitch contours to capture emotional and neurological cues.
- **Achieved 83%+ accuracy** on real time data in autism detection and improved depression classification by 25% using continual learning across varied datasets and speakers.
- **Applied adaptive learning techniques** to improve generalization and handle speaker variability in real-world audio samples.

PROJECTS

Patent Project – Descriptive Answer Sheet Evaluation System)| AI/ML and Full Stack | April 2024 – Present | [Link](#) | [GitHub](#)

- Designed and built an Agentic AI-driven handwritten answer sheet evaluation system using OCR, LaTeX parsing, and fine-tuned OpenAI LLMs integrating LangGraph; tested on 250+ students over 3 months (April-June).
- Developed and deployed rubric management, stepwise logical grading, natural language feedback, secure report storage, and admin dashboards. Approved by the college and currently tested on 1000–1500 students across B.A. LL.B. (Law), B. Tech (CS & core), Health Sciences, and Commerce.

AI-Powered Mock Investor Pitching Platform | Web Development and AI/ML | January 2025 | [Link](#) | [GitHub](#)

- Built and deployed a highly scalable backend system using OpenAI Whisper, Google Gemini, and ElevenLabs/Google TTS, now acquired by VertxAI for internal use.
- Enabled real-time audio interaction, handling audio processing, session management, and multi-persona investor simulations and mapping investors to suitable founders.

SharpView | Low-Light Image Enhancement | Java | UPES | November 2024 | [GitHub](#)

- Developed a tool in **Java** for enhancing low-light and degraded images using advanced algorithms such as **Gamma Correction, Grey World Algorithm, and Gaussian Smoothing**, enabling quality improvements for real-time processing with an accuracy of 87%.

- Integrated **Flask** backend with responsive frontend and engaging web interface for real-time image enhancements.

Audio Processing: Autism and Depression Detection AI Research | Nov 2023 – Apr 2024 | [GitHub](#)

- Built deep learning systems using Whisper, LLaMA3, CNN, RNN, LSTM, and Transformers for audio-based mental health diagnostics.
- Achieved 90%+ accuracy in autism detection and a 25% improvement in depression detection using continual learning across multiple datasets.

EDUCATION

University of Petroleum and Energy Studies (UPES)

Dehradun, India

B.Tech in Computer Science(Honours) with specialisation in Artificial Intelligence & Machine Learning: **8.9 CGPA** **June 2022 - June 2026**

St Conrad's Inter College | 12th ISC PCM with Computer Science: **92.5%** | 10th ICSE Mathematics: **95.3%**

Agra, India

SKILL SUMMARY

- **Languages:** Java, Python, JavaScript, C, TypeScript
- **Machine Learning:** Generative AI, Agentic AI, NLP, Computer Vision, Audio Processing, Deep Learning, Continual Learning, MPC.
- **Web Development:** MongoDB, ReactJS, NodeJS, AngularJS, Tailwind CSS
- **Frameworks/Tools:** n8n, Pidata(agno), Git, GitHub, Google Colab, Jupyter, Postman, Docker
- **Soft Skills:** Problem-Solving, Leadership, Teamwork, People Management, Communication

ACHIEVEMENTS

- Drafted **2** patents for the **DASES (Descriptive Answer Sheet Evaluation System)** project under university innovation initiatives.
- Awarded SHODH Grant of ₹49,000 for developing LexiBot: AI-Powered Legal Assistant to aid in consumer, traffic, and women harassment cases (2025–26).
- **DevOps Bootcamp:** Secured **first** place in the **DevOps Bootcamp conducted by INT Israel**, competing against 200+ participants.
- **Smart India Hackathon:** Shortlisted for the Smart India Hackathon, one of India's premier innovation competitions, organized by the Government of India **for two consecutive years (2024 & 2025)**.

Links

- **Founder and Technical Head | CCS | Full- Time** | [Link](#)
- **LexiBot** | AI-powered legal assistant providing guidance for consumer, traffic, and women harassment cases. | CSS | [Link](#)
- **Resume-JD Matcher** | CCS | 500+ active users | [Link](#) | [GitHub](#)
- **Internship Evaluator** UPES Dashboard Prototype| CCS | 7000+ uses (Scheduled) | [Link](#)
- Investor Founder Mapping Website | VertxAI Labs | Internship | [Link](#) | [GitHub](#)
- IIIT Delhi and Ulster University Machine Learning Contributions | [GitHub](#)
- SharpView | Low-Light Image Enhancement | UPES | [GitHub](#)